EchoScribe: Evidence-Linked, Independently Verified Notes for Therapy and Mentoring

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***Abstract*—Therapists and academic mentors spend a large share of their working time writing structured session records. Language models can draft these records from a recording, but two problems block practical use: a draft may state things the session never contained, and a single cloud model is a single point of failure for quota, outage and data-governance reasons. We present EchoScribe, an open system that turns a recorded two-person session into a SOAP (therapy) or GROW (mentoring) note in which every sentence cites the transcript turns that support it. An independent model, from another vendor when one is configured and otherwise a different model, checks each sentence against its cited turns; when too many sentences are unsupported, a verification-gated cascade regenerates the note on another model and keeps the better draft. A second model also screens independently for risk, and disagreement is surfaced for review instead of being resolved silently. Every processing role is served by an embedded multi-provider router with routing strategies, per-key exponential cooldown and a provider-wide breaker. On the 40 test dialogues of ACI-Bench, 80.6% of generated sentences were judged supported by the independent verifier, which detected 98% of planted errors with a 9% false-alarm rate. In fault-injection simulations the last-known-good strategy cut 95th-percentile latency from 31.7 s to 9.1 s while keeping success at 100.00%. We report speech recognition, speaker attribution and role identification on public data and state plainly what remains unvalidated.**

***Index Terms*—clinical documentation, hallucination detection, speaker diarization, large language models, LLM routing, SOAP notes, GROW coaching**

I. Introduction

Writing session notes is one of the most time-consuming non-clinical tasks of mental health work and of academic mentoring. A SOAP note records the client's account, the clinician's observations, an assessment and a plan [1]; a GROW note records a mentee's goal, current reality, options and way forward [2]. Documentation extends the working day well beyond the session [3], and ambient AI scribes that draft notes from recorded encounters are now in clinical use and under randomized evaluation [4], [5].

Two gaps limit what such systems can be trusted with. First, a drafted note may contain statements the conversation never supported [6], [7], and the reviewer has no fast way to tell which sentence rests on what. Second, published scribes depend on one model from one vendor, so a quota, an outage, a model retirement or a change in an institution's approved-vendor list stops documentation altogether. Neither gap is addressed for counselling or mentoring conversations, where prior work concentrates on medical visits [8]–[10].

EchoScribe addresses both gaps in one pipeline. Its contributions are:

1) **Evidence-linked notes with independent verification.** The writer must cite turn numbers for every sentence; a model other than the writer, preferably from another vendor, labels each sentence as supported, partially supported or unsupported; a verification-gated cascade regenerates notes that fail a threshold on another model (Section III-C).
2) **An embedded multi-provider router** that serves every processing role from a *combo* of targets with a routing strategy, per-key exponential cooldown, a provider-wide breaker and request-scoped error handling, adapted from the open-source OmniRoute gateway [11] (Section III-A).
3) **Two-party speaker attribution without enrolment**: local pyannote diarization of the original recording with two fallback paths, followed by role resolution from eight conversation statistics and a model check (Section III-B).
4) **A quantitative evaluation** on public data of the router under injected faults, note quality and grounding on ACI-Bench, verifier accuracy on planted errors, role identification, and speech recognition and speaker attribution under noise (Section IV).

II. Related Work

A. Ambient scribes and note generation

Clinical note generation from doctor–patient dialogue has been benchmarked on ACI-Bench [10] and in the MEDIQA-Chat shared tasks [12], [13]. Adapted language models can match or exceed experts on clinical summarisation [14], and

open models have been tuned for expert-level notes with physician reader studies [15]. Evaluation frameworks for ambient scribes now combine computational metrics, reviewer assessment and simulation [8], and deployed systems such as LAOS report recognition and physician review results [9]. For mental health, Adhikary et al. benchmarked language models on summarising counselling sessions [16]. None of these studies produces mentoring (GROW) notes, links each note sentence to its evidence, or remains available when its model vendor fails.

B. Hallucination, grounding and verification

Hallucination is a recurring failure of generated text [6]; Asgari et al. propose a clinical safety framework for annotating hallucinations and omissions in medical summaries [7]. Generic remedies include generating text with citations [17], scoring atomic facts against a source [18], sampling-based self-consistency checks [19] and self-verification plans [20]. EchoScribe combines citation-constrained generation with sentence-level verification by an *independent model*, preferably from another vendor so that the checker does not share the writer's training data or failure modes, and uses the verdicts both for the reviewer interface and to trigger regeneration.

C. Speaker diarization, enhancement and recognition

Diarization combines speech activity detection, speaker embeddings and clustering [21]. ECAPA-TDNN embeddings [22] grouped by spectral clustering [23] remain a strong baseline, while end-to-end systems such as pyannote's powerset model [24] and Sortformer [25] learn segmentation directly; DiarizationLM post-corrects speaker labels with a language model [26], and WhisperX aligns recognition with diarization [27]. Zolensky et al. study role identification in clinical conversations [28]. SepFormer [29], distributed in SpeechBrain [30] and trained on WHAM! [31], is one of many enhancement models benchmarked in the DNS challenges [32]; enhancement artefacts can also *hurt* recognition [33], which we test rather than assume.

D. Routing across language models

RouteLLM learns to route queries between a strong and a weak model [34], and FrugalGPT cascades models to reduce cost [35]. Open gateways such as OmniRoute [11] provide combo routing with cooldowns in front of many vendors. EchoScribe embeds this mechanism per processing role inside the application and adds verification as a routing signal: a note is escalated not when a call fails but when its content fails verification.

III. System Design

Fig. 1 shows the pipeline. A browser client records or uploads audio; an Express.js server coordinates every model call; Supabase stores records under per-user row-level security, with therapy and mentoring records separated by a mode column that every query must state. Five roles call language or speech models: STT, DIARIZE, ROLE_LABEL, NOTE_SYNTH and VERIFY.

A. Embedded multi-provider router

Each role is served by a *combo*: an ordered list of targets ⟨vendor, model, keys⟩ and a routing strategy. Vendors are thin adapters exposing `chatCompletion`, `transcribeAudioBuffer` or `diarizeAudioBuffer`; Gemini [36], Groq-hosted gpt-oss-120b [37] and Whisper [38], Deepgram and a local pyannote service ship as adapters. Combos come from the user's settings, from environment variables, or from a reference configuration. Several keys for one model form a pool whose keys take turns. The strategy orders targets per request: *priority* (declared order; the next target is used only when every key of the current one is cooling down or retired, or the model itself fails), *round-robin*, *random*, *strict-random* (a shuffled deck), *least-used*, and *lkgp* (the last target that succeeded goes first).

Failures are classified. A key that fails authentication is retired; a rate-limited or failing key enters an exponential cooldown of $5 \cdot 2^{k-1}$ s (capped at 5 min) after its k-th consecutive failure, reset on success; a vendor with 15 failures within 30 min is skipped for 10 min (2 in 5 min for the local service); an HTTP 400 is treated as specific to the request and moves on without cooling the key. These mechanisms and their constants are adapted from OmniRoute [11]. When every target fails, the API returns HTTP 503 with the list of attempts. Keys are kept in the user's browser by default; a user may opt to sync an individual key to the account, where it is stored encrypted with AES-256-GCM and never returned to the client.

B. Speech processing and role resolution

The recording is not denoised: SepFormer enhancement, our first design, raised word error rates (Section IV), as reported for other front ends [33]. DIARIZE runs pyannote community-1 [24], [39] locally on the original 16 kHz recording, on a GPU when present, with two speakers and non-overlapping turns; it falls back to cloud audio diarization, then to splitting the transcript with a language model. STT transcribes the original recording; when the recogniser returns word timestamps (Whisper, Deepgram), each word goes to the turn that contains it [27], otherwise text is spread over turns by duration. A denoised copy (FFmpeg) is made only for listening again.

Roles are resolved without voice enrolment. For each speaker the system computes eight statistics: speaking time, word count and mean turn length, questions, professional vocabulary, guiding phrases, personal and emotional phrases, number of turns, and who spoke first. ROLE_LABEL receives the transcript with these statistics and must assign exactly one professional and one client; invalid answers move to the next target, and a deterministic default is used only if every target fails.

C. Evidence-linked notes and independent verification

The labelled transcript is numbered into turns $T_1 \dots T_n$. NOTE_SYNTH must end every sentence of the note sections, and of the risk notes, with the turns that support it ("…poor

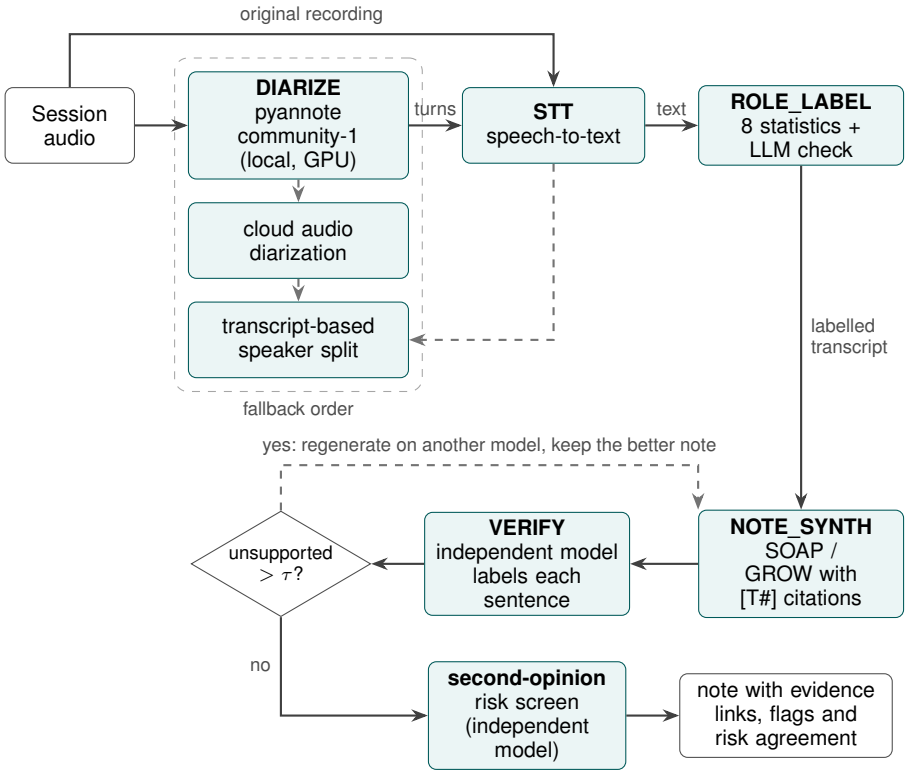


Fig. 1. EchoScribe pipeline. Shaded boxes are processing roles, each served by a combo of the embedded router (Section III-A). Speaker separation falls back from local pyannote to cloud audio diarization to a transcript-based split (dashed). The note writer cites transcript turns; an independent model (another vendor, else another model) verifies each sentence, and notes over the unsupported threshold τ are regenerated on another model. A second model screens independently for risk.

sleep for two weeks [T4, T6]"), and must write "Not discussed" rather than an uncited sentence. Citations are parsed into per-sentence evidence and removed from the displayed text. VERIFY then receives each sentence with only its cited turns and labels it *supported*, *partial* (adds a detail the turns lack) or *unsupported*. The router orders verifiers as another vendor first, then another model of the writer's vendor, and never uses the writer's own model, so verification is cross-vendor whenever a second vendor is configured and cross-model otherwise.

If the unsupported fraction exceeds $\tau = 0.2$, the note is regenerated with the writer's model moved to the end of the combo, re-verified, and the draft with fewer unsupported sentences is kept. The interface shows each sentence's evidence on hover and marks partial and unsupported sentences for the reviewer (Fig. 2).

D. Risk screen with a second opinion

In therapy mode the writer also screens for suicidal ideation and self-harm; in mentoring mode, for burnout and severe distress. A second model, chosen like the verifier, repeats the screen independently and cites its evidence turns. The flag is raised if either model raises it, favouring sensitivity, and disagreement is shown as "models disagree — review the transcript". The screen is a model judgement, not a validated instrument such as the C-SSRS or PHQ-9 [40], [41]; language models are known to vary in suicide-risk judgements [42], and the interface states that the clinician remains responsible for risk assessment.

IV. Evaluation

All experiments are scripted in the public repository (`eval/`) and every number below is generated from their result files.

A. Setup

Data. Note generation, verification and role identification use the 40 test dialogues of ACI-Bench (test1, CC BY 4.0) [10], whose doctor–patient visits have reference notes; no public corpus of therapy or mentoring sessions with reference notes exists. Speech experiments use 12 simulated two-person conversations (12.5 min) built from LibriSpeech test-clean [43]: ten alternating utterances from a mixed-gender speaker pair with 0.3–0.9 s gaps, exact reference turns, and three conditions, clean and with babble plus pink noise at 5 dB and 0 dB SNR. **Models.** Gemini models [36] are used in rotation across three keys; gpt-oss-120b [37] and Whisper large-v3 [38] are served by Groq. The model that served each request is logged. **Metrics.** ROUGE [44] and BERTScore [45] against reference notes; WER after text normalisation; DER with a 0.25 s collar [46]; and cpWER, the speaker-attributed WER of CHiME-6 [47], which scores every speaker-separation

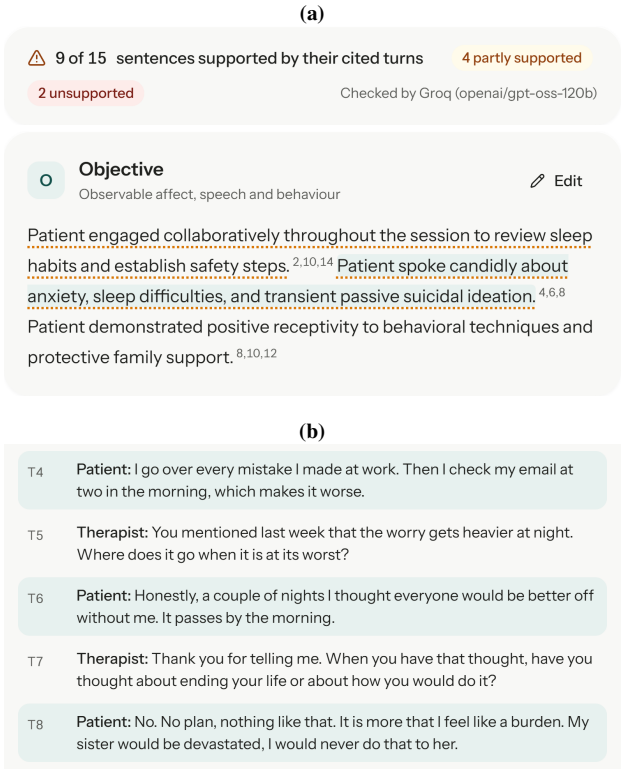


Fig. 2. EchoScribe's session summary on a role-played therapy session (real pipeline output; transcript invented). (a) The verification bar counts sentences the independent verifier (here gpt-oss-120b, checking a Gemini-written note) judged supported, partly supported or unsupported; in the Objective section, dotted underlines mark partly supported sentences and superscripts give cited turns. (b) Hovering the second Objective sentence highlights the turns it cites (T4, T6, T8) in the transcript.

path including those without timings. Intervals are Wilson 95%.

B. Router under injected faults

We drive the router's own code with simulated vendors on a virtual clock: one hour of Poisson traffic at 10 requests/min against a preferred vendor (two keys), a fallback vendor (two keys) and a second fallback, in three scenarios: a per-key quota below the offered load (*quota*); a 15-minute outage of the preferred vendor (*outage*); and a revoked key, a quota and a fallback that times out 20% of the time (*mixed*). We compare walking the chain on every request with no memory, the fixed 60 s suspension of our earlier design, and the router with three strategies (5 seeds each, Table I).

No policy changes success much where a healthy fallback exists, so the differences lie in wasted calls and latency. The last-known-good strategy stops returning to a vendor that keeps timing out and reduces 95th-percentile latency in the mixed scenario from 31.7 s to 9.1 s; round-robin wastes the fewest calls and spreads load most evenly (Jain index [48] 0.90 under quota pressure). Exponential cooldown alone is not uniformly better than a fixed suspension: under a persistent per-minute quota its 5 s first step re-probes an exhausted key (0.227 against 0.098 failed calls per request), while the fixed rule can suspend every option at once and lost requests in the mixed scenario (98.14% success). Priority keeps the clinician's preferred model at the cost of probing; LKGP minimises latency but can stay on a fallback model after the preferred one recovers.

TABLE I
ROUTER UNDER INJECTED FAULTS (MEAN OF 5 SEEDS): SUCCESS RATE, FAILED PROVIDER CALLS PER REQUEST, AND 95TH-PERCENTILE LATENCY (S).

	Quota			Outage			Mixed		
Policy	Succ.	Fail	p95	Succ.	Fail	p95	Succ.	Fail	p95
No memory	99.97%	0.476	2.4	99.97%	0.291	2.4	99.93%	0.497	31.8
Fixed 60 s	100.00%	0.098	2.3	99.89%	0.087	2.4	98.14%	0.147	31.5
Priority	99.97%	0.227	2.4	99.97%	0.062	2.3	99.86%	0.258	31.7
Round-robin	100.00%	0.014	2.7	99.93%	0.058	2.7	100.00%	0.092	31.4
LKGP	99.97%	0.033	1.9	99.93%	0.040	2.4	100.00%	0.097	9.1

TABLE II
NOTES ON ACI-BENCH TEST1: OVERLAP WITH REFERENCE NOTES AND INDEPENDENT VERIFICATION OF EVERY SENTENCE.

Writer (verifier)	Gemini (gpt-oss)	gpt-oss (Gemini)
Notes	40	40
ROUGE-1 / 2 / L	45.4 / 16.9 / 42.5	47.0 / 14.4 / 42.8
BERTScore F1	48.0	44.1
Sentences per note	18.2	16.8
Sentences with citations	95.6%	97.3%
Supported / partial	83.9% / 9.2%	77.0% / 16.3%
Unsupported	6.9%	6.7%
Notes over gate (τ=0.2)	1	4
Planted errors caught: number	95%	95%
negation	100%	100%
borrowed sentence	100%	100%
False alarms on unchanged	7%	12%

C. Evidence-linked notes and independent verification

Each of the two writers produced a SOAP note with turn citations for the 40 test dialogues, and a model other than the writer verified every sentence (Table II). Because the Groq free tier's daily token cap was reached during the runs, verification of Gemini-written notes was cross-vendor for 15 notes and cross-model (another Gemini model) for 25; for gpt-oss-written notes it was cross-vendor for 40 and cross-model for 0. Notes that the writer's own model had checked during a quota fallback were re-verified by another model, and one gpt-oss note that came back empty was regenerated, so all notes are counted. Citation coverage was 95.6% (Gemini) and 97.3% (gpt-oss-120b); 83.9% and 77.0% of sentences were judged fully supported and 6.9% and 6.7% unsupported. With the gate at $\tau = 0.2$, 1 of 40 Gemini-first notes would have been regenerated. ROUGE against ACI-Bench reference notes measures overlap with a differently structured visit note and is reported for comparability, not as a quality ceiling.

Does the verifier catch errors? Model-judged support is only useful if the judge detects real errors. We planted three kinds of error into sentences the verifier had accepted: a changed number, a flipped negation, and a sentence borrowed from another dialogue's note with the original citations; other

TABLE III
SIMULATED TWO-PERSON CONVERSATIONS (12 × 3 CONDITIONS, %). TOP: RECOGNITION WITH AND WITHOUT SEPFORMER DENOISING, AND SPEAKER SEPARATION (DER). BOTTOM: SPEAKER-ATTRIBUTED CPWER OF EACH PATH; THE FIRST ROW IS THE SUBMITTED DESIGN, THE BOLD ROW THE CURRENT ONE.

	Clean	5 dB	0 dB
WER Whisper, original / denoised	2.1 / 2.9	13.5 / 28.9	36.0 / 70.9
WER Gemini, original / denoised	5.9 / 4.0	16.9 / 26.9	45.5 / 65.0
DER pyannote / SpeechBrain	9.3 / 23.1	12.9 / 25.1	22.6 / 24.9
SepFormer + SpeechBrain + Gemini	43.4	48.8	77.9
pyannote + Whisper, by duration	21.4	47.1	68.8
pyannote + Whisper, by word time	**2.6**	**19.7**	**53.2**
SpeechBrain + Whisper, by word time	21.0	27.1	46.9
Cloud audio diarization (Gemini)	17.8	37.1	59.3
Transcript-based split	38.4	34.9	62.0

accepted sentences were kept as controls. Across 109 planted errors the verifier flagged 98% as partial or unsupported, with 9% false alarms on 129 unchanged sentences (Table II, bottom). Controls were sentences this verifier had already accepted, so the false-alarm rate measures consistency rather than precision on fresh sentences.

D. Role identification

With doctor and patient labels replaced by random speaker identifiers, the eight statistics alone named the clinician correctly in 38 of 40 dialogues (95.0%), the model alone in 37 (92.5%), and their combination, as deployed, in 39 (97.5%). Only 2 dialogues had the patient speaking more than the doctor, too few to test the case the statistics are meant for; this is part of the role-play protocol below.

E. Speech recognition and speaker attribution

Table III reports the speech results. Denoising before recognition, as in our first design, raised word error rates under noise for both recognisers (for Whisper at 0 dB from 36.0% to 70.9%), so the deployed pipeline transcribes the original recording. pyannote separated speakers with 9.3% DER on clean audio against 23.1% for SpeechBrain clustering. Speaker attribution depends as much on placing words as on finding turns: spreading the transcript over pyannote's turns by duration gave 21.4% cpWER, assigning each word to the turn containing its Whisper timestamp gave 2.6%, and at 5 dB 19.7% against 37.1% for cloud audio diarization. At 0 dB pyannote's DER rises to 22.6% and SpeechBrain clustering with word timestamps attributes slightly better (46.9% against 53.2%).

Per minute of audio, pyannote took 2.1 s on a laptop GPU (RTX 4050) and 22.0 s on its CPU, against 78.2 s for the first design's SepFormer enhancement with SpeechBrain diarization; Whisper on Groq took 10.2 s and Gemini speech-to-text 67.3 s.

V. DISCUSSION AND LIMITATIONS

What the results support. Citation-constrained writing kept almost every sentence attached to evidence, and an independent vendor caught most planted factual errors, so sentence-level flags give a reviewer a concrete place to look instead of an unexplained confidence score [49]. The router results show that resilience comes from remembering failures, and that the choice of strategy is a trade-off between staying on the preferred model and minimising latency, not a single best setting. Role identification without enrolment was accurate on two-party clinical dialogue, and the combination of statistics and model was more accurate than either alone.

Deployment. EchoScribe runs as a local application: the server, the speech service and the database client run on the clinician's machine, and only the configured model vendors receive audio or text. Keys stay in the browser unless the user chooses to sync one. Sessions are not limited in length; long recordings are compressed to mono Opus before they reach a cloud model, speaker separation runs locally (on a GPU when present), and timeouts scale with duration. Because model access is per role and per vendor, an institution can restrict processing to vendors its data-governance policy approves, and speaker separation never leaves the machine unless the local service is unavailable.

Limitations. (i) ACI-Bench contains medical visits, not therapy or mentoring sessions; no public corpus of counselling sessions with reference notes exists, and GROW notes were not evaluated quantitatively. (ii) Support labels come from a model; the planted-error study bounds how often it misses errors of three kinds, but subtler omissions and misattributions need human review [7]. (iii) Speech results come from simulated conversations of read speech without overlap; spontaneous, overlapping clinical speech will be harder. (iv) Results depend on specific model versions that vendors retire; we log the serving model of every request. (v) The risk screen has not been validated against clinician labels. The protocol released with the code covers these gaps: blinded ratings of evidence-linked against single-pass notes by clinicians and mentors, annotated role-played sessions for real-speech WER, DER and role accuracy, and a clinician-labelled vignette probe of the risk screen.

VI. CONCLUSION

EchoScribe produces SOAP and GROW notes from recorded two-person sessions in which every sentence points to the turns that support it and is checked by a model other than its writer, with a verification-gated cascade, a two-model risk screen that surfaces disagreement, and an embedded router that keeps every processing role available through quota exhaustion, outages and revoked keys. On public data the notes were almost fully cited, the independent verifier caught most planted errors, role identification needed no enrolment, and the router's last-known-good strategy cut tail latency under compound faults. The code, evaluation scripts and human-evaluation protocol are available at https://github.com/Aravind-Krishnan-S/ECHOSCRIBE.

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